Communication lower bounds from the asymptotic spectrum of tensors

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Editorial scope. Counted author principal result: thm:asymptspectrum. Other correspondence lemmas, special cases and group applications are uncounted context. Exact tensor-invariant and promise restrictions are retained.

Allowed standard prerequisites (editorial).

- Standard tensor products, matrix-multiplication tensor, known independent-set constructions and triangle-removal theorem. Exact imported statements and full original bibliography retained; all local promise/coloring/block constructions and invariant arguments included.

1 Introduction

Number on the forehead (NOF) communication complexity was introduced by Chandra, Furst and Lipton [CFL83] as a variant of Yao's model of communication complexity. Here, we consider k players, each of them having one of k inputs, but instead of providing the players with their inputs, we imagine them having their inputs written on their foreheads — in particular each player can observe all inputs *except* for their own. Similar to the standard communication complexity, the players wish to exchange as little communication as possible, in order to jointly compute some function of their inputs.

As it turns out this innocuous alteration has deep mathematical consequences. For example, being able to prove lower bounds for specific NOF problems with $k = \Theta(\log^c n)$ players would imply explicit circuit complexity lower bounds [HG90, BT94].

Moreover, even if we consider only three players, the theory of NOF communication complexity is simultaneously extremely deep, and frustratingly lacking. Its depth comes in part from a number of fascinating and perhaps surprising connections between natural questions in NOF communication complexity and well-studied problems in Ramsey Theory, extremal combinatorics and additive combinatorics. For example, already in the seminal paper [CFL83], a surprisingly efficient deterministic protocol was provided for the NOF problem of checking whether $x+y+z = 0$ in $\mathbb{Z}_N$, where $x, y, z \in \mathbb{Z}_N$ are the inputs on the players' foreheads (we will call this problem $\textsc{Eval}_{\mathbb{Z}_N}$ from now on). The protocol uses just $\mathcal{O}(\sqrt{\log N})$ communication, and makes use of Behrand's construction of a relatively dense set without three-terms arithmetic progression [Beh46]. On the

other hand, in the same paper they showed an $\omega(1)$ lower bound for the $\textsc{Eval}_{\mathbb{Z}_N}$ problem using the Hales–Jewett theorem — a deep theorem in Ramsey theory. Since then, connections with the Ruzsa–Szemerédi problem, Corner-Free Sets and other objects from extremal combinatorics have been discovered, frequently leading to state-of-the-art constructions [LPS19, AS20, AMS12, LS21, CFTZ22].

At the same time, perhaps because of these connections, many seemingly simple questions about NOF communication complexity remain unresolved, despite the fact that their number-in-hand variants are very easy to answer. For example, completely resolving the communication complexity of $\textsc{Eval}_{\mathbb{Z}_N}$ requires a significantly better understanding of the correct quantitative dependency in the Roth's theorem (that large density subsets of $[N]$ contain a 3-terms arithmetic progression), and seems to be far out of reach of current techniques.

Similarly, it is still open how to find an explicit 3-party NOF problem which can be solved with $\mathcal{O}(1)$ randomized communication complexity (assuming shared randomness), but which requires $\Omega(n)$ deterministic communication, even though this is a relatively simple property of Equality for two-party communication, and even though, with a non-constructive argument, it is possible to show that such NOF problems exist [DP08]. A natural candidate for such a problem is checking whether $x + y + z = 0$ in $\mathbb{F}_3^n$ (or, more generally, if $xyz = 1$ in a large power G^n of some fixed group G, not necessarily abelian — we will call those problems $\textsc{Eval}_{G^n}$), but proving a NOF communication complexity lower bound has been so far elusive.

Problems like $\textsc{Eval}_{G^n}$, in which for any pair of inputs to two of the players, there is exactly one input to the remaining player for which they should accept, will be called *permutation problems*. These problems always have $\mathcal{O}(1)$ randomized communication complexity, so they are natural candidates to try to prove $\Omega(n)$ deterministic communication lower bounds. Indeed, a number of recent works have studied permutation problems toward this goal [BDPW10, CFTZ22].

The construction [CFL83] of an efficient protocol for $\textsc{Eval}_{\mathbb{Z}_N}$ can be interpreted as a statement that existence of a large set $S \subset G$ without three terms arithmetic progressions in an abelian group G can be used to provide an efficient protocol for $\textsc{Eval}_G$; by contrapositive, a lower bound for the communication complexity of $\textsc{Eval}_G$ would imply an upper bound for the size of any such set S. This latter statement for the group $\mathbb{F}_3^n$ used to be known as a *cap-set conjecture* for few decades, and has famously been positively resolved in 2016 [EG16, CLP17, Tao16] — the largest set $S \subset \mathbb{F}_3^n$ without three terms arithmetic progressions has size $\mathcal{O}(c^n)$ for some $c < 3$. This leads to a natural question: can the techniques used to prove the cap-set theorem be generalized to prove a linear lower bound for the communication complexity of $\textsc{Eval}_{\mathbb{F}_3^n}$?

Christandl et al. [CFTZ22] recently identified a barrier against a direct applications of such techniques — they used a method called *combinatorial degenration* to prove that a *subrank* of a specific tensor associated with the communication problem $\textsc{Eval}_{G^n}$ for an abelian group G is near-maximal — whereas an upper bound on this subrank would imply a desired lower-bound for communication complexity.

1.1 Our contributions

In this paper, we study three-player communication through the lens of *promise number-in-hand problems*. For an alphabet Σ, a promise problem (I, P) consists

of two subsets $I \subset P \subset \Sigma \times \Sigma \times \Sigma$. In this problem, the three players are given as input $a, b, c \in \Sigma$, respectively, with the promise that $(a, b, c) \in P$, and their goal is to determine whether $(a, b, c) \in I$. For instance, usual number-in-hand communication corresponds to this model where $P = \Sigma \times \Sigma \times \Sigma$.

NOF problems can also be viewed as promise number-in-hand problems. Each player is given as an input a pair $(a_i, b_i) \in \Sigma^2$ for $i \in \{1, 2, 3\}$, and their inputs are guaranteed to satisfy that $a_2 = b_1, a_3 = b_2$ and $a_1 = b_3$. (For instance, one thinks of a_2 and b_1 as the input on the head of player 3, so we are promised that players 1 and 2 see the same value there.) Write $P_{\text{NOF}} \subset \Sigma^2 \times \Sigma^2 \times \Sigma^2$ to denote the inputs that satisfy that promise.

A key idea we pursue is to interpret I and P, not just as subsets of inputs, but also as order-3 (3-dimensional) $\{0, 1\}$-valued tensors. For example, letting $N = |\Sigma|$, we can represent P_{NOF} as a $\{0, 1\}$-valued tensor $P_{\text{NOF}} \in \mathbb{F}^{N^2 \times N^2 \times N^2}$, with the tensor entry being one whenever the corresponding tuple is a legal input.

As it turns out, P_{NOF} is *exactly* the matrix multiplication tensor, the same tensor which is extensively studied in the theory of fast matrix multiplication algorithms! We leverage this extremely surprising, if only formal connection, to transfer ideas which have been developed in studying fast matrix multiplication to give new insights into NOF communication complexity, and conversely, to use results proved originally about NOF communication complexity and obtain new alternative proofs of consequential properties of the matrix multiplication tensor.

We use this connection to prove the following results.

Communication Protocols from the Laser Method We use the Laser method, the technique introduced by Strassen and used to design the best-known matrix multiplication algorithms [Str86, CW87], as a tool to design NOF communication protocols. For any three NOF problems I, J, K, let $I \otimes J \otimes K$ ("their product") denote the NOF problem of solving the three simultaneously, i.e., given one input to each, determining whether all three are accepting inputs. Using the Laser method, we show that for any three permutation NOF problems I, J, K with n-bit inputs, the product $I \otimes J \otimes K$ (which has $3n$-bit inputs) can be solved using only n bits of communication deterministically.

For example, for any permutation NOF problem I with n-bit inputs, one can consider the problem of, given N simultaneous inputs to I, determining if they are all accepting inputs. A consequence of our protocol is that this can be solved with $\frac{1}{3}nN + \mathcal{O}(n)$ bits of communication deterministically. This implies constructions of surprisingly large corner-free sets for powers of arbitrary abelian groups G.

By contrast, we show using a counting argument that a random NOF permutation problem I with n-bit inputs has requires deterministic communication complexity $\geq n/3 - \mathcal{O}(1)$. It was not previously clear to what extent this is tight; our protocol shows that this bound can at least be achieved for any permutation problem which is itself the product of three permutation problems.

Our protocol should also be contrasted with the proof by probabilistic method in [CFTZ22], which showed that every NOF permutation problem with n-bit inputs has a protocol with communication complexity $n/2 + \mathcal{O}(1)$, improving in turn upon the trivial n upper bound.

The Zeroing Out Subrank of Matrix Multiplication A quantity which appears frequently in the theory of matrix multiplication algorithms is the *zeroing out subrank* of matrix multiplication, denoted $Q_{zo}(\langle n, n, n\rangle)$. This is the size of the largest identity tensor (3-dimensional analogue of the identity matrix) which can be achieved as a zeroing out of the tensor $\langle n, n, n\rangle$ for multiplying two $n \times n$ matrices. Strassen [Str86] famously proved that $Q_{zo}(\langle n, n, n\rangle) \geq n^{2-o(1)}$, which is roughly as large as possible (since $\langle n, n, n\rangle$ is an $n^2 \times n^2 \times n^2$ tensor), and used it as part of the Laser method to design a faster algorithm for matrix multiplication[1]. Recent *barrier* results have also used this bound to rule out certain approaches to designing faster matrix multiplication algorithms [AW21, Alm21, CVZ21].

In the bound $Q_{zo}(\langle n, n, n\rangle) \geq n^{2-o(1)}$, it is natural to ask what is hidden by the $o(1)$. For instance, it was recently shown that the 'border subrank' of $\langle n, n, n\rangle$ is $\lceil \frac{3}{4}n^2 \rceil$ [KMZ20]; is it possible that we also have $Q_{zo}(\langle n, n, n\rangle) \geq \Omega(n^2)$? This could improve low-order terms in some applications.

In fact, we prove that this is not possible, and that $Q_{zo}(\langle n, n, n\rangle) < n^2/\omega(1)$. To prove this, we build off of the work of [LPS19], who showed a NOF communication lower bound, that any NOF permutation problem requires $\omega(1)$ deterministic communication. Using our connection between NOF communication and matrix multiplication, we are able to modify their proof to get our upper bound on $Q_{zo}(\langle n, n, n\rangle)$.

Interestingly, the prior work [LPS19] proved their communication lower bound using a connection with the Ruzsa–Szemerédi problem from extremal combinatorics. Not only does our new proof show a connection between $Q_{zo}(\langle n, n, n\rangle)$ and the Ruzsa–Szemerédi problem, but we are in fact able to show that the two are *equivalent*! More precisely, it is impossible to determine the precise asymptotics of $Q_{zo}(\langle n, n, n\rangle)$ without also resolving the Ruzsa–Szemerédi problem. This is interesting in contrast with the 'border subrank' which has been precisely determined [KMZ20].

Finally, using the connection with NOF communication, we also provide an alternative proof of Strassen's lower bound $Q_{zo}(\langle n, n, n\rangle) \geq n^{2-o(1)}$, deducing it as a corollary of the existence of the efficient deterministic NOF protocol for $\textsc{Eval}_{\mathbb{Z}_N}$.

Slice-Rank Methods Since their introduction [CFL83], NOF problems I have been associated with a corresponding 3-hypergraph H whose chromatic number exactly characterizes the deterministic communication complexity of I. One recent powerful approach for bounding the chromatic number of such a hypergraph is the 'slice rank' method, which was introduced to resolve the cap-set conjecture [CLP17, EG16]. Roughly speaking, if one can show that the slice rank of the adjacency tensor of H is not too large, then this implies the chromatic number of H is not too small, and hence a communication lower bound.

Since then, a few successful techniques have been introduced to bound the

[1] See, e.g., [Blä13, Section 8]. Bounds used in algorithms are sometimes described as bounds on a 'monomial degeneration subrank' or 'combinatorial degeneration subrank', but these are in turn ultimately used to bound the zeroing out subrank via [Bin05]. Note that bounds on more general notions like 'border subrank' or 'subrank' would not suffice in these algorithms, since the subrank decomposition is applied only to the 'outer structure' of a tensor in the Laser method [Str86].

slice ranks of many different tensors [Tao16, BCC$^+$17]. Recent work [CFTZ21] asked whether these techniques could give linear deterministic NOF communication lower bounds. However, they showed that, unfortunately, it is impossible to use such techniques to prove communication lower bounds for any EVAL group problem. They proved this by defining the corresponding adjacency tensor and calculating that it has maximal 'asymptotic subrank' and hence large slice rank.

We generalize their result and show that slice-rank methods cannot be used to prove a linear deterministic communication lower bound for *any* NOF permutation problem (not just an EVAL group problem). To show this, we observe that for any NOF permutation problem, the corresponding adjacency tensor is actually (a permutation of) the matrix multiplication tensor P_{NOF}. Since the matrix multiplication tensor is known to have maximal asymptotic subrank [Str86], it follows that the adjacency tensor never has small enough slice rank. In particular, this implies that in hindsight, the tensor whose asymptotic subrank was computed in the prior work [CFTZ21] is exactly the matrix multiplication tensor.

By contrast, we use slice-rank methods to prove linear lower bounds for other promise problems. The key is that this method works whenever the *promise tensor* has low slice-rank. For example, consider the promise problem $\text{EQ}_{\mathbb{F}_3^N}$, where parties receive inputs $x, y, z \in \mathbb{F}_3^N$ with a promise that $x + y + z = 0$, and they wish to decide whether $x = y$. We prove that this problem requires $\Omega(N)$ communication. This may seem like it should be straightforward to prove ("how could the third player's input $z = -(x + y)$ possibly help the first two players to test for equality?"), but we prove that the problem is actually *equivalent* to the cap-set problem (which was a long-open conjecture).

General Rank Methods and the Asymptotic Spectrum of Tensors Slice-rank is one way to measure the 'complexity' of a tensor, but there are many others, including other notions of tensor rank, and more generally a spectrum of different measures which is well-studied in algebraic complexity and tensor combinatorics called the 'asymptotic spectrum of tensors' [Str86, WZ22]. Let r be any of these measures (the reader may want to focus on the case when r is the tensor rank). We generalize the slice-rank method and prove that for any promise problem (I, P), its deterministic communication complexity is at least $\log_2(r(I)/r(P))$. Hence, finding a measure r which is larger for the problem I than the promise P leads to a communication lower bound. (This is why, for slice rank, the fact that the promise P_{NOF} has maximal slice rank means that a lower bound cannot be proved in this way.)

This already has intriguing consequences when r is tensor rank R.

For number-in-hand communication, we know that the promise tensor P_{NIH} is the all-1s tensor which has rank $R(P_{NIH}) = 1$. Hence, for any number-in-hand problem I, we get a communication lower bound of $\log_2(R(I))$. This is analogous to the standard upper bound in two-party deterministic communication compleixty by the log of the rank of the communication matrix. Moreover, any $2^n \times 2^n \times 2^n$ tensor I (corresponding to an n-bit input problem) which is not 'degenerate' in some way has rank at least 2^n, which explains why nearly all number-in-hand communication problems require linear communication.

For NOF communication for n-bit inputs, as we've discussed, the promise tensor P_{NOF} is the matrix multiplication tensor. Determining the rank of ma-

trix multiplication is the *central question* when designing matrix multiplication algorithms; the matrix multiplication exponent ω is defined exactly such that the rank of P_{NOF} is $R(P_{NOF}) = (2^n)^{\omega+o(1)}$. Hence, for any problem I with rank $R(I) \geq (2^n)^c$, we get a communication lower bound of $(c - \omega - o(1)) \cdot n$. The fact that ω appears negated in this lower bound is intriguing: this means that designing faster matrix multiplication algorithms, and hence lowering the known upper bound on ω, leads to an improved communication lower bound for such problems I.

Matrix Multiplication as a Communication Problem Finally, we consider how other promise problems can be used to shed further light on NOF communication lower bounds. Let I be any n-bit NOF problem, let $N = 2^n$, and let $T_{(\mathbb{Z}/N\mathbb{Z})^2}$ denote the structure tensor of the group $(\mathbb{Z}/N\mathbb{Z})^2$, i.e., the tensor

$$T_{(\mathbb{Z}/N\mathbb{Z})^2} = \sum_{a,b,c,d \in [N]} x_{(a,b)} y_{(c,d)} z_{(a+c \pmod N, b+d \pmod N)}.$$

We observe that $P_{NOF} \subset T_{(\mathbb{Z}/N\mathbb{Z})^2}$ (after appropriately permuting variables), and so it is well-defined to consider three different promise problems using these three tensors: (I, P_{NOF}), $(P_{NOF}, T_{(\mathbb{Z}/N\mathbb{Z})^2})$, and $(I, T_{(\mathbb{Z}/N\mathbb{Z})^2})$. Moreover, a simple triangle inequality-style argument shows that their communication complexities are related via:

$$\mathcal{CC}(I, P_{NOF}) \geq \mathcal{CC}(I, T_{(\mathbb{Z}/N\mathbb{Z})^2}) - \mathcal{CC}(P_{NOF}, T_{(\mathbb{Z}/N\mathbb{Z})^2}).$$

On the left-hand side is exactly the deterministic NOF complexity of the problem I that we would like to prove a lower bound for. On the right-hand side, $(I, T_{(\mathbb{Z}/N\mathbb{Z})^2})$ is a similar problem to the original NOF problem of I, but with a weaker promise, and so it should be easier to prove communication lower bounds for. What about $\mathcal{CC}(P_{NOF}, T_{(\mathbb{Z}/N\mathbb{Z})^2})$? (Recall that in this problem, we're given as input a term from the tensor $T_{(\mathbb{Z}/N\mathbb{Z})^2}$, and our goal is to determine whether or not it is a valid NOF input from P_{NOF}.)

For this problem, we give a protocol with communication $\leq n$ (the trivial bound would be $2n$). Moreover, we prove a communication lower bound of $\geq (\omega - 2)n$ (this follows from the rank-based approach discussed above). So, further improvements to the communication protocol actually *require* fast matrix multiplication (since they would imply an upper bound on ω via this inequality). If it were the case that $\omega = 2$ (which is popularly conjectured), then it is plausible that this problem actually has sublinear communication complexity. In that case, in order to prove a linear lower bound for the NOF problem (I, P_{NOF}), it would suffice to prove a linear lower bound for the seemingly-harder problem $(I, T_{(\mathbb{Z}/N\mathbb{Z})^2})$. In summary, if we could design a faster matrix multiplication algorithm achieving $\omega = 2$, then this may be a promising approach to proving NOF communication lower bounds.

2 Tensor Preliminaries

In this paper we will relate communication problems to related tensors and their properties. We will focus on order-3 tensors (also known as 3-dimensional

tensors). For a field $\mathbb{F}$ and three positive integers A, B, C, we define $\mathbb{F}^{A\times B\times C}$ to be the set of tensors

$$T = \sum_{a\in A}\sum_{b\in B}\sum_{c\in C} T_{a,b,c}x_a y_b z_c$$

for coefficients $T_{a,b,c} \in \mathbb{F}$ (where x_a, y_b, z_c are formal variables). When the specific sets are not important, we will often write $\mathbb{F}^{|A|\times|B|\times|C|}$ instead of $\mathbb{F}^{A\times B\times C}$.

T is $\{0,1\}$-valued if, for all a, b, c, we have $T_{a,b,c} \in \{0,1\}$. Such tensors are in bijection with subsets of $A \times B \times C$. All tensors considered in this paper are assumed to be $\{0,1\}$-valued unless stated otherwise.

The field $\mathbb{F}$ that one works over will be primarily important to us for the following definition. The (not necessarily $\{0,1\}$-valued) tensor T has rank 1 if it can be written as

$$T = \left(\sum_{a\in A}\alpha_a x_a\right)\left(\sum_{b\in B}\beta_b y_b\right)\left(\sum_{c\in C}\gamma_c z_c\right)$$

for coefficients $\alpha_a, \beta_b, \gamma_c \in \mathbb{F}$. More generally, the rank of the (not necessarily $\{0,1\}$-valued) tensor T, denoted $R(T)$, is the minimum number of tensors of rank 1 which sum to T.

Rank is critical to the definition of the matrix multiplication exponent ω. For $n, m, p \in \mathbb{N}$, we write $\langle n, m, p\rangle \in \mathbb{F}^{n^2\times n^2\times n^2}$ to denote the matrix multiplication tensor

$$\langle n, m, p\rangle = \sum_{i=1}^{n}\sum_{j=1}^{m}\sum_{k=1}^{p} x_{i,j}y_{j,k}z_{k,i}.$$

The exponent ω is defined as

$$\omega := \inf\{t \mid R(\langle n, n, n\rangle) \leq n^t \text{ for some } n \in \mathbb{N}\}.$$

Indeed, it is known that for any $\varepsilon > 0$, matrix multiplication of $n\times n$ matrices can be performed with an arithmetic circuit of size $\mathcal{O}(n^{\omega+\varepsilon})$, and conversely, that any arithmetic circuit for matrix multiplication can be converted into a tensor rank upper bound which yields the same operation count in this way [Str73].

We say T is an identity tensor if, for all $a \in A$, there is at most one pair $(b, c) \in B \times C$ such that $T_{a,b,c} \neq 0$, and similarly for all $b \in B$ and for all $c \in C$. Often, when $A = B = C$, one specifically thinks of the identity tensor of size k as a tensor $\sum_{a\in S} x_a y_a z_a$ for some subsets $S \subset A$ of size $|S| = k$. To be clear that this is sufficient but not necessary (as one could in general permute the indices), we will refer to permutations of identity tensors.

If $T, T' \in \mathbb{F}^{A\times B\times C}$, we say T' is a zeroing out of T if there are subsets $A' \subset A, B' \subset B, C' \subset C$ such that $T'_{a,b,c}$ is equal to $T_{a,b,c}$ whenever $a \in A', b \in B', c \in C'$, and $T'_{a,b,c} = 0$ otherwise. In other words, we are setting coefficients outside of A', B', C' to zero. We will sometimes write $T' = T|_{A',B',C'}$.

The zeroing out subrank of T, denoted $Q_{zo}(T)$, is the maximum k such that T has a zeroing out to a permutation of an identity tensor of size k. This more combinatorial variant on the notion of 'subrank' (which uses 'restrictions' rather than zeroing outs) will be useful a number of times.

3 Promise problems and colored tensors

Definition 1. *A promise problem over alphabet Σ is a pair of subsets $I \subset P \subset \Sigma \times \Sigma \times \Sigma$, where P is a set of allowed inputs (the* promise*), and I is the subset of accepting inputs (the* problem*).*

To study those objects, we will introduce a notion of *colored tensors*. A colored tensor is a pair (I, T_P), where $T_P \in \mathbb{F}^{\Sigma\times\Sigma\times\Sigma}$ is a tensor and $I \subset \mathrm{supp}(T_P)$ is a subset of its non-zero terms. (We imagine those terms to have a special color, and in what follows we will call them *green* terms). 3-party promise number in hand problems over the alphabet Σ are therefore in direct correspondence with $\{0,1\}$-valued colored tensors over $\mathbb{F}^{\Sigma\times\Sigma\times\Sigma}$, where the non-zero terms of the tensor T corresponds to the set of allowed inputs, and the subset I of green terms corresponds to accepting inputs.

Definition 2. *For any field $\mathbb{F}$ we can define the* communication tensor *of a promise problem (I, P) as a colored tensor (I, T_P), with $T_P \in \mathbb{F}^{\Sigma\times\Sigma\times\Sigma}$ given by*

$$T_P := \sum_{(i,j,k)\in P} x_i y_j z_k.$$

We will say that T_P is the promise tensor *of the problem (I, P).*

We will sometimes abuse notation, and use the same symbol to denote a $\{0,1\}$-valued tensor $P \in \mathbb{F}^{A\times B\times C}$ and its support $\mathrm{supp}(P) \subset A \times B \times C$ whenever it is clear from context. All tensors we discuss in this paper are $\{0,1\}$-valued and for the vast majority of the discussion the underlying field is irrelevant.

Communication Complexity Model We use $\mathcal{CC}(I, P)$ to denote the deterministic communication complexity of the promise problem (I, P). To make this concrete, we consider a shared blackboard model: here a protocol is given by a triple of functions, one for each player, encoding what each player should append to the blackboard in each round of communication, given the current state of the blackboard, and their private input. Players write their communication on the blackboard in cyclic order; in each stage each player can either append something to the blackboard, accept the current instance, or reject the current instance. A specific instance is accepted if *all players* accept it (based on their own input, and the final state of the blackboard). A protocol solves a promise problem (I, P) if it accepts all the inputs from I and rejects all other inputs from $P \setminus I$. (Since P is the promise, the protocol need not have any particular behavior on inputs outside of P.)

Asymptotic communication complexity For a promise problem (I, P) we use $(I, P)^{\otimes n}$ to denote a promise problem in which parties are given n valid instances of the original promise problem (I, P), and wish to determine whether *all* of those instances simultaneously are accepting.

More formally, $(I, P)^{\otimes n}$ is the promise problem with the promise $P^{\otimes n} \subset (\Sigma^n) \times (\Sigma^n) \times (\Sigma^n)$, i.e., for $A, B, C \in \Sigma^n$, the triple of sequences (A, B, C) is in $P^{\otimes n}$ if and only if we have $(A_i, B_i, C_i) \in P$ for all $i \in [n]$. The set of accepting instances $I^{\otimes n}$ is defined analogously. This operation corresponds to the Kronecker product of the associated communication tensors.

Definition 3. *For tensors $I^{(1)} \subset T^{(1)} \subset \mathbb{F}^{A_1 \times B_1 \times C_1}$ and $I^{(2)} \subset T^{(2)} \subset \mathbb{F}^{A_2 \times B_2 \times C_2}$ We define the Kronecker product of colored tensors $(I^{(1)}, T^{(1)}) \otimes (I^{(2)}, T^{(2)})$ as a colored tensor (I, T) in $\mathbb{F}^{A_1 A_2 \times B_1 B_2 \times C_1 C_1}$, where $T = T_1 \otimes T_2$ satisfies*

$$T_{(a_1,a_2),(b_1,b_2),(c_1,c_2)} = T^{(1)}_{a_1,b_1,c_1} \cdot T^{(2)}_{a_2,b_2,c_2},$$

and $I = \{(a_1, a_2), (b_1, b_2), (c_1, c_2) : (a_i, b_i, c_i) \in I_i \text{ for } i \in \{1, 2\}\}$.

We remark that all tensors considered in this work are order-3 tensors (also known as 3-dimensional tensors), and we will use symbol $\otimes$ exclusively to denote the Kronecker product of tensors (as opposed to tensor product).

We will define the asymptotic communication complexity of the problem P as

$$\underline{\mathcal{CC}}(I, P) := \lim_{n \to \infty} \frac{\mathcal{CC}((I, P)^{\otimes n})}{n}.$$

Permutation problems We pay special attention to a particular class of communication problems, which we will call *injection problems* — as we will see later on, the deterministic communication complexity of injection problems have very simple combinatorial interpretation, and they all have protocols with constant randomized communication (under public randomness).

Definition 4 (Injection problem)**.** *Let $\pi_1, \pi_2, \pi_3 : \Sigma^3 \to \Sigma$ be projections onto first, second, and third coordinate respectively.*

We say that (I, P) (where $I \subset P \subset \Sigma^3$) is an injection problem*, if the tensor $T_I := \sum_{(i,j,k) \in I} x_i y_j z_k$ restricted to the coordinates appearing in I (i.e. $(T_I)|_{\pi_1(I),\pi_2(I),\pi_3(I)}$) is a permutation of an identity tensor.*

That is: for every valid input of the $i \in \Sigma$ of the first player, there is at most one pair j, k, such that $(i, j, k) \in I$, and analogously for the second and the third player.

Definition 5 (Permutation problem)**.** *We say that (I, P) is a permutation problem if (I, P) is an injection problem and moreover $|I| = \Sigma$.*

Note that if (I, P) is an injection (resp. permutation) problem, then $(I, P)^{\otimes n}$ also is an injection (permutation) problem.

3.1 Number on the forehead problems as promise-number-in-hand problems

We can treat a NOF problem $I \subset \Sigma \times \Sigma \times \Sigma$ as a promise problem over the alphabet $\Gamma := \Sigma \times \Sigma$, where the promise $P_{\text{NOF}} \subset \Gamma^3$ is given by the following promise-tensor

$$P_{\text{NOF}} := \sum_{i,j,k \in \Sigma} x_{i,j} y_{j,k} z_{k,i}.$$

A key observation that we will explore in this paper is that P_{NOF} is exactly the matrix multiplication tensor $P = \langle N, N, N \rangle$ where $N = |\Sigma|$.

The set of accepting instances $I \subset \Gamma \times \Gamma \times \Gamma$ can be bijectively mapped into a subset $\tilde{I} \subset \langle N, N, N \rangle$ of non-zero terms of the promise tensor, by $\tilde{I} := \{((x, y), (y, z), (z, x)) : (x, y, z) \in I\}$.

Having interpreted number-on-the-forehead problems as number-in-hand problems with promise $\langle N, N, N \rangle$, we observe that our more general definition of the

permutation problem agrees with the definition of the permutation problem in [LPS19] for NOF problems — which are in turn are equivalent to Latin squares of size $N \times N$.

An important family of examples of NOF permutation problems are EVAL problems induced by an abelian group G. Specifically, for any abelian group G, let EVAL_G be the problem defined as follows. Players have $x, y, z \in G$ on their foreheads, and they want to decide whether $x + y + z = 0$. We can represent the problem by its set of accepting instances $I \subset G^3$ defined by $I = \{(x, y, z) : x + y + z = 0\}$.

The observation that all NOF permutation problems admit an efficient randomized protocol, generalizes to injection problems regardless of the promise, using a constant communication randomized protocol for two-players EQUALITY problem.

Fact 6. *Every injection problem has a $O(1)$ randomized protocol with shared randomness, regardless of the promise.*

Proof. First player, upon receiving his input x can find unique y_1, z_1, such that $(x, y_1, z_1) \in I$. He can then proceed by using the randomized two-player protocol for EQUALITY to check whether $y_1 = y$, and $z_1 = z$, where y, z are inputs of the second and third player respectively. □

What can we say about the deterministic communication complexity of promise injection problems? It turns out that we can express it equivalently as a question about the smallest proper coloring of the communication tensor (I, P).

Definition 7. *Consider a colored tensor $T = (I, P)$, We say that a subset of its green terms $S \subset I$ is an independent set if S is a permutation of an identity tensor and the restriction $T|_{\pi_1(S),\pi_2(S),\pi_3(S)}$ is equal to S. We use $\alpha(T)$ to denote the size of largest independent set in the colored tensor T.*

If T is an ordinary tensor, we use $\alpha(T)$ to denote the size of the largest independent set in the colored tensor $(\text{supp}(T), T)$ (i.e., we treat all its terms as green). This coincides with the usual notion of zero-out subrank of a tensor.

Note that if $T = (I, P)$ is an injection problem, $S \subset I$ is an independent set if and only if all non-zero terms in the restriction $T|_{\pi_1(S),\pi_2(S),\pi_3(S)}$ are in I — that is, if we denote by $S_i := \pi_i(S)$ then set S is independent if and only if

$$\text{supp}(T) \cap (\pi_1(S_1)^{-1} \times \pi_2(S_2)^{-1} \times \pi_3(S_3)^{-1}) = S.$$

With a definition of an independent set, we can define a proper coloring and the chromatic number of a colored tensor in a natural way.

Definition 8. *For a colored tensor $T := (I, P)$ we say that $\tau : I \to [k]$ is a proper coloring if for all colors $c \in [k]$, the set $\tau^{-1}(c)$ is independent.*

We denote by $\chi(T)$ the chromatic number *of tensor T — the smallest k, such that there is a proper coloring $\tau : I \to [k]$. Again, if T is an ordinary tensor, we use a shorthand $\chi(T) := \chi(\text{supp}(T), T)$.*

The proof of the following characterization follows exactly the original proof in [CFL83], just phrased in a slightly more general language of promise problems instead of specifically Number on the Forehead problems.

Theorem 9. *The deterministic communication complexity of any promise injection problem (I, P) is equal to $\lceil \log \chi(I, P) \rceil$.*

Proof. Given a coloring $\tau : I \to [k]$, we will describe a communication protocol with complexity $\lceil \log k \rceil$. The first player, with an input x, looks at the only $y^{(1)}, z^{(1)}$, s.t. $(x, y^{(1)}, z^{(1)}) \in I$, and writes down the color $\tau(x, y^{(1)}, z^{(1)})$. Similarly, the second player, with an input y, compares the written color with the color of $(x^{(2)}, y, z^{(2)}) \in I$, and analogously the third player. Clearly, if (x, y, z) was an accepting instance, all three of the triples in the consideration would be the same $(x^{(2)} = x^{(3)} = x, y^{(1)} = y^{(3)} = y, z^{(1)} = z^{(2)} = z)$, and therefore the colors under consideration would agree — the protocol will correctly accept such an instance.

On the other hand, if all $(x, y^{(1)}, z^{(1)}), (x^{(2)}, y, z^{(2)}), (x^{(3)}, y^{(3)}, z)$ have the same color c, let us take the independent set $S := \tau^{-1}(c) \subset I$, then $x \in \pi_1(S), y \in \pi_2(S), z \in \pi_3(S)$, and since $(x, y, z) \in P$ and S is an independent set, we must have $(x, y, z) \in I$ — i.e. the instance was accepting. This completes the proof that a coloring $\tau : I \to [k]$ implies existence of an efficient protocol.

The converse is similar: given a protocol for a problem (I, P) with communication at most k, we can use the transcript of the communication on each input $(x, y, z) \in I$ as a color for the term. This yields a mapping $\tau : I \to [2^k]$, and all we need to check is that it is a proper coloring. Indeed, let us chose an arbitrary color c, and take a set $S = \tau^{-1}(c)$. If there was $(x, y, z) \in P - I$, such that $(x, y, z) \in \pi_1(S) \times \pi_2(S) \times \pi_3(S)$, the communication transcript on the (allowed) input (x, y, z) would have been the same as on accepting inputs $(x, y^{(1)}, z^{(1)}), (x^{(2)}, y, z^{(2)}), (x^{(3)}, y^{(3)}, z)$, and therefore each player would have accepted it — contradicting correctness of the protocol, since $(x, y, z) \notin I$. □

3.2 Comparison with standard bounds for the NOF communication complexity

What we discussed so far is a slightly different perspective on the standard way of characterizing NOF communication complexity (already present in [CFL83]). Classically, with NOF permutation problem $I \subset \Gamma \times \Gamma \times \Gamma$, one associated a directed 3-hypergraph H, and the logarithm of the chromatic number of this hypergraph is equal to the deterministic communication complexity of the associated problem.

We will show that in fact the adjacency tensor of the aforementioned hypergraph H is just a permutation of the promise tensor P_{NOF} (i.e., the matrix multiplication tensor), and the permutation is uniquely determined by the problem I — it corresponds to permuting variables y_i and z_i in a way such that I is exactly on the diagonal of the tensor.

This provides a generalization, and another perspective on the barrier against using a slice-rank or any other bounds on the subrank of the adjacency tensor of H as a way to show a lower bound for communication complexity of H.

Specifically, note that any independent set $S \subset V(H)$ induces a zeroing-out of the adjacency tensor of the hypergraph H to a large identity subtensor (given by restriction to indices in $S \times S \times S$ — note that in the construction of the hypergraph H and its adjacency tensor, we include self-loops (t, t, t) for each $t \in V(H)$). One could then hope to show a lower bound on the chromatic number of the hypergraph H by upper bounding the size of the largest

independent set, which in turn can be upper bounded by subrank of adjacency sub-tensor. Finally, it is known that subrank of any tensor is always bounded by its slice-rank, and a few successful techniques for controlling slice-ranks of particular tensors of interest have been developed [Tao16, BCC$^+$17].

This, apparently promising avenue for showing lower bounds for specific NOF permutation problems unfortunately has to fail. In [CFTZ22] it was shown that similar techniques as those used by Strassen to prove lower bound on the asymptotic subrank of the matrix multiplication tensor, can be used to lower bound the asymptotic subrank of the adjacency tensor for any EVAL group problem — it is maximally large. We observe that this is not a coincidence: in fact all adjacency tensors constructed this way are just permutations of the matrix-multiplication tensor, so the lower bound for asymptotic subrank of the matrix multiplication tensor directly applies — not only for problems arising as EVAL$_G$, but more generally for all number-on-the-forehead permutation problems.

Definition 10 (Communication Hypergraph [CFL83])**.** *For a permutation NOF problem $I \subset \Gamma \times \Gamma \times \Gamma$, we define its communication hypergraph $H(I)$ to be an order-3 hypergraph, with vertices $V(H(I)) = I$ — set of all accepting instances, and hyperedges constructed in the following way: for all $(x, y, z) \in \Gamma \times \Gamma \times \Gamma$, let $T_1 := (x, y, z') \in I$ be the only triple in I with agreeing with (x, y, z) in the first two coordinates, and similarly $T_2 := (x', y, z) \in I, T_3 := (x, y', z) \in I$.*

Then $E(H(I)) = \{(T_1, T_2, T_3) : (x, y, z) \in \Gamma^3\}$.

Fact 11. *The adjacency tensor of the hypergraph $H(I)$ is a permutation of the matrix multiplication tensor $\langle|\Gamma|, |\Gamma|, |\Gamma|\rangle$.*

Moreover if we consider a set of green terms $\tilde{I} \subset \langle|\Gamma|, |\Gamma|, |\Gamma|\rangle$ given by $\tilde{I} = \{((x, y), (y, z), (z, x)) : (x, y, z) \in I\}$ in the colored communication tensor for the problem I, this set corresponds under the aforementioned permutation exactly to the set of diagonal terms $\{(t, t, t) : t \in I\}$ of the adjacency tensor of $H(I)$.

Proof. Consider a map $\pi_{12} : I \to \Gamma \times \Gamma$, given by $\pi_{12}(x, y, z) = (x, y)$, and similarly π_{23}, π_{13}. Note that those three bijections between I and $\Gamma \times \Gamma$. Applying those bijections to three axes of the adjacency tensor A of the hypergraph $H(I)$, we see that an image of any hyperedge (T_1, T_2, T_3) is in the Theorem 10 is $(\pi_{12}(T_1), \pi_{23}(T_2), \pi_{13}(T_3)) = ((x, y), (y, z), (x, z))$ — a non-zero term of the matrix multiplication tensor $\langle|\Gamma|, |\Gamma|, |\Gamma|\rangle$. All the non-zero terms of the matrix multiplication tensor can be obtained this way: since, by construction of the communication hypergraph, any term $((x, y), (y, z), (x, z))$ can be lifted to an hyperedge $(T_1, T_2, T_3) \in H(I)$.

Finally, the statement about the image of the diagonal follows directly from the definition of permutations π_{12}, π_{23} and π_{13} and construction of the hypergraph H. □

Corollary 12. *For any NOF permutation problem, the adjacency tensor of its communication hypergraph has maximal asymptotic subrank.*

3.3 Lower bounding the communication complexity of random NOF permutation problem

In this section we will show that a random NOF permutation problem over the alphabet $[N] \times [N] \times [N]$, where $N = 2^n$ requires $\geq n/3 - \mathcal{O}(1)$ deterministic

communication. This is a simple counting argument, similar in spirit to the one used in [DP08]. Since the NOF permutation problems over $[N] \times [N] \times [N]$ are bijective with Latin squares of size $N \times N$, known upper and lower bounds for the number of Latin squares can be used in our proof.

The following is a consequence of the fact that any Latin square can be constructed by repeatedly taking a perfect matching in a regular bipartite graph, and using standard bounds on the number of perfect matchings in a k-regular bipartite graph.

Theorem 13 ([VLW01])**.** *Total number $\#P$ of NOF permutation problems over $[N] \times [N] \times [N]$ satisfies*

$$N^{N^2} e^{-N^2} \leq \#P \leq N^{N^2}.$$

With this in hand, we can prove

Theorem 14. *For most of NOF permutation problems (I, P), over the alphabet $[N]$ where $N = 2^n$, we have $\mathcal{CC}(I, P) \geq \frac{n}{3} - \mathcal{O}(1)$.*

Proof. Let (I, P) be a NOF permutation problem over $[N] \times [N] \times [N]$ with communication complexity d. By Theorem 9 communication complexity is equivalent to the existence of proper coloring $\kappa : I \to \{0,1\}^d$. Since I is a permutation problem, for any $x \in [N] \times [N]$, there is a unique $\pi_1^{-1}(x) \in I \subset ([N] \times [N])^3$ (where $\pi_1 : ([N] \times [N])^3 \to [N] \times [N]$ is the projection onto first coordinate). We can define $\kappa_1 : [N] \times [N] \to \{0,1\}^d$ as $\kappa_1(x) := \kappa(\pi_1^{-1}(x))$. Analogously, we define κ_2 and κ_3.

We observe that the triple of functions $(\kappa_1, \kappa_2, \kappa_3)$ uniquely determines I. Indeed, for $(a, b, c) \in P$ to check if $(a, b, c) \in I \iff \kappa_1(a) = \kappa_2(b) = \kappa_3(c)$. Therefore, the number of bits to specify a problem with communication complexity d (i.e. the logarithm of the number of such problems) is at most $3dN^2$ — to specify a function $\kappa : [N] \times [N] \to \{0,1\}^d$ we need dN^2 bits.

On the other hand, using the lower bound of Theorem 13, we need $N^2 \log N - \mathcal{O}(N^2)$ bits to write down arbitrary NOF permutation problem. Therefore if $d \leq (\log N)/3 - \mathcal{O}(1)$, most of the problems do not have a protocol with communication complexity d. □

3.4 Background on independence number and coloring of matrix multiplication tensor

The following statement asserting that matrix-multiplication tensor has almost maximal zero-out subrank is an important insight in the study of matrix multiplication. It has been originally proven by Strassen in [Str87], and has been used as a technical step in the design of some of the fast matrix multiplication algorithms. Here we will use it as a tool in the proof of Theorem 20. On the other hand, in Section 3.6 we will discuss how, using our connection between number on the foreheads protocol and structural properties of the matrix multiplication tensor, the following statement can be easily deduced from the known results in the NOF communication complexity on efficient protocols for some permutation problems.

Theorem 15 ([Str87])**.** *There is $I \subset \mathrm{supp}\langle N, N, N\rangle$, such that $|I| \geq N^{2-o(1)}$, and the restriction $\langle N, N, N\rangle|_{\pi_1(I),\pi_2(I),\pi_3(I)}$ is a permutation of the identity tensor, i.e. $Q_{zo}(\langle N, N, N\rangle) \geq N^{2-o(1)}$.*

Equivalently, in the notation introduced in this work,

$$\alpha(\langle N, N, N\rangle) \geq N^{2-o(1)}.$$

The following is a standard technique of lifting the large independent set to a small coloring in symmetric combinatorial objects. We include the proof for completeness.

Definition 16 (Automorphism group)**.** *For a colored tensor $T = (I, P)$, we define $\mathrm{Aut}(T)$ to be the set of all triples of permutations $(\sigma_1, \sigma_2, \sigma_3)$ such that $\sigma(I) = I$, and moreover for all (x, y, z) we have $T_{x,y,z} = T_{\sigma_1(x),\sigma_2(y),\sigma_3(z)}$.*

Lemma 17. *For a colored tensor $T = (I, P)$ if for any two green elements $a, b \in I$, there is a $\sigma \in \mathrm{Aut}(T)$, s.t. $\sigma(a) = b$, then*

$$\chi(T) \leq \frac{|I|}{\alpha(T)} \log |I|.$$

Proof. Let $S \subset I$ be an independent set, and $\sigma \sim \mathrm{Aut}(T)$ a uniformly random element of the automorphism group. Then clearly $\sigma(S)$ is also an independent set. We claim that for any $x \in I$, we have $\Pr(x \in \sigma(S)) = \frac{|S|}{|I|}$. Indeed, let $S = \{s_1, s_2, \ldots s_m\}$. The subset $T \subset \mathrm{Aut}(T)$ of permutations σ such that $\sigma(s_1) = x$ is exactly a coset of a subgroup H in $\mathrm{Aut}(T)$ where H is a set of permutations fixing x, i.e. $H = \{\sigma : \sigma \in \mathrm{Aut}(T), \sigma(x) = x\}$. Number of such cosets is exactly $|I|$, since we assumed that x can be mapped to any other $s \in I$ by some permutation in $\mathrm{Aut}(T)$ (and no other element, by the property of automorphism group that each $\sigma \in \mathrm{Aut}(T)$ preserves green terms: $\sigma(I) = I$). Since all cosets of a given subgroup are disjoint and have the same size, we get $\Pr(\sigma(s_1) = x) = 1/|I|$). Therefore

$$\Pr(x \in \sigma(S)) = \sum_{i \leq |S|} \Pr(x = \sigma(s_i)) = |S|/|I|.$$

Finally, if we consider $K = C(|I| \log |S|)/|S|$ independent random permutations $\sigma_1, \sigma_2, \ldots, \sigma_m \in \mathrm{Aut}(T)$, we wish to say that with positive probability for every element $x \in I$ it is covered by at least one set $\sigma_i(S)$. Indeed, in expectations, each of those is covered by $C \log |S|$ such sets, and applying a standard Chernoff and union bound argument, we reach the conclusion.

Having a covering of I by K independents sets, we can easily find a coloring $\tau : I \to [K]$ mapping an elemnet $x \in I$ to the smallest $i \in [K]$ such that $x \in \sigma_i(S)$. □

Corollary 18. *The chromatic number of the matrix multiplication tensor satisfies*

$$\chi(\langle N, N, N\rangle) \leq N^{1+o(1)}.$$

We shall also need the following standard fact on the Kronecker products of matrix multiplication tensors.

Fact 19 (see, e.g., [Blä13, Page 24])**.** *The Krnoecker product of matrix multiplication tensors satisfies*

$$\langle N_1, M_1, P_1\rangle \otimes \langle N_2, M_2, P_2\rangle = \langle N_1 N_2, M_1 M_2, P_1 P_1\rangle$$

3.5 Laser method gives a non-trivial asymptotic protocol for all NOF permutation problems

In this section we will use the celebrated Laser method, introduced to design fast matrix multiplication algorithms [Str86, CW87], to prove the following surprising upper bound on the asymptotic NOF communication complexity of the third player of arbitrary permutation problem.

Theorem 20. *For every triple of NOF permutation problems I, J, K, each over $\{0,1\}^n \times \{0,1\}^n \times \{0,1\}^n$, we have*

$$\mathcal{CC}(I \otimes J \otimes K) \leq (1 + o(1))n.$$

(Note that the trivial bound is $\mathcal{CC}(I \otimes J \otimes K) \leq 3n$.)

In particular for any NOF permutation problem P over $\{0,1\}^n \times \{0,1\}^n \times \{0,1\}^n$, we have

$$\underline{\mathcal{CC}}(P) \leq (1 + o(1))\frac{n}{3}.$$

We define the *block colored tensor* to be a colored tensor (I, T) over $A \times B \times C$ together with a collection of partitions $A = A_1 \cup \ldots A_p$, $B = B_1 \cup \ldots \cup B_q$, and $C = C_1 \cup \ldots C_q$.

The *outer structure* of a block tensor T, is a tensor $\mathrm{Out}(T)$ over $[p] \times [q] \times [r]$ with $\mathrm{Out}(T)_{i,j,k} = 1$ if and only if the restriction $T|_{A_i,B_j,C_k} \neq 0$ (and $\mathrm{Out}(T)_{i,j,k} = 0$ otherwise). The *inner structure* $\mathcal{I}(T)$ of a block tensor T is a collection of (colored) tensors $T|_{A_i,B_j,C_k}$. The notions of outer and inner structure feature prominently in the Laser method; see e.g., [Blä13, Section 8].

Lemma 21. *For a block-colored tensor T, we have*

$$\chi(T) \leq \chi(\mathrm{Out}(T)) \cdot \max_{T' \in \mathcal{I}(T)} \chi(T').$$

Proof. This is immediate — to find a proper coloring for the set of green terms I of a block tensor T, we can map an element $x \in I$ to (c_1, c_2), where c_1 is a color assigned to the block in which x lies in the coloring of $\mathrm{Out}(T)$, and c_2 is a color assigned to x in the coloring of the block itself.

We need to show that each color class is an independent set. With an induced partition of A, B, C, this amounts to observing that for any block tensor for which $\mathrm{Out}(T)$ is is a permutation of the identity tensor, as well as all tensors in $\mathcal{I}(T)$, the entire tensor also is a permutation of the identity tensor. □

Fact 22. *Let T^1, T^2 be a pair of block-colored tensor. Then $T := T^1 \otimes T^2$ is a block colored tensor, with the outer structure $\mathrm{Out}(T) = \mathrm{Out}(T^1) \otimes \mathrm{Out}(T^2)$, and the inner structure $\mathcal{I}(T) \subset \mathcal{I}(T^1) \otimes \mathcal{I}(T^2)$ (Here, for two collections of tensors $\mathcal{A}$ and $\mathcal{B}$ we use $\mathcal{A} \otimes \mathcal{B} := \{a \otimes b : a \in \mathcal{A}, b \in \mathcal{B}\}$.)*

Proof. Directly follows from the definition. □

Fact 23. *For any pair of colored tensors T_1, T_2, we have $\chi(T_1 \otimes T_2) \leq \chi(T_1)\chi(T_2)$.*

Proof. Given two colorings $\kappa_1 : I_1 \to [k_1]$ and $\kappa_2 : I_2 \to [k_2]$ we can construct a coloring of the Kronecker product $T_1 \otimes T_2$ using pairs $[k_1] \times [k_2]$, by mapping a term $((i_1, i_2), (j_1, j_2), (k_1, k_2)) \in I_1 \otimes I_2$ to a pair $(\kappa_1(i_1, j_1, k_1), \kappa_2(i_2, j_2, k_2))$. The fact that it is a proper coloring follows directly from the fact that Kronecker product of two identity tensors is again an identity tensor. □

Lemma 24. *There is a triple of partitions $\mathcal{A}, \mathcal{B}, \mathcal{C}$ such that for any NOF permutation-problem (I, P_{NOF}), the block-colored tensor $T := (I, P_{NOF}, \mathcal{A}, \mathcal{B}, \mathcal{C})$ induced by those partitions satisfies the following conditions*

- *The outer structure* $\mathrm{Out}(T) = \langle N, 1, 1\rangle$.
- *Every colored tensor in the inner structure* $(Y, T') \in \mathcal{I}(T)$ *has* $T' = \langle 1, N, N\rangle$ *and* $\chi(Y, T') = 1$.

Proof. Consider partitions $\mathcal{A} := (A_1, \ldots, A_n)$ where $A_t := \{(i, t) : i \in [N]\}$, similarly $\mathcal{B} := (B_1, \ldots B_N)$ where $B_t = \{(t, i) : i \in [N]\}$, and finally $\mathcal{C} := (C) := ([N] \times [N])$ is a trivial partition with a single part.

First note that with this partition of the matrix multiplication tensor P, the block $A_{t_1} \times B_{t_2} \times C$ is non-empty if and only if $t_1 = t_2$,

$$\mathrm{Out}(T) = \sum_t x_{1,t} y_{t,1} z_{1,1},$$

is the matrix multiplication tensor $\langle N, 1, 1\rangle$.

Moreover, let us consider an arbitrary non-empty block $A_t \times B_t \times C$. Tensor T restricted to this block is

$$T|_{A_t \times B_t \times C} = \sum_{j,k} x_{j,t} y_{t,k} z_{k,j},$$

which is exactly a $\langle 1, N, N\rangle$ matrix multiplication tensor.

Note now that the projection $\pi_3 : A_t \times B_t \times C \to C$ induces a bijection between C and $Y := \mathrm{supp}(T|_{A_t \times B_t \times C})$. If we look at $S := I \cap Y$, we wish to argue that S is an independent set in $T|_{A_t \times B_t \times C}$. Indeed, denoting $S_i := \pi_i(S)$, since (I, P_{NOF}) is a permutation problem, all we need to argue is that

$$\pi_1^{-1}(S_1) \times \pi_2^{-1}(S_2) \times \pi_3^{-1}(S_3) \cap Y = S.$$

But since π_3 is bijection, we already have $(A_t \times B_t \times \pi_3^{-1}(S_3)) \cap Y = S$. □

We are now ready to prove Theorem 20.

Proof of Theorem 20. For permutation NOF problems I, J, K, according to Theorem 24 we can chose a block-colored tensor structure T_I, T_J, T_K, such that $\mathrm{Out}(T_I) = \langle N, 1, 1\rangle$, $\mathrm{Out}(T_J) = \langle 1, N, 1\rangle$ and $\mathrm{Out}(T_K) = \langle 1, 1, N\rangle$.

By Theorem 22 and Theorem 19, the block-colored tensor $T = T_I \otimes T_J \otimes T_K$ satisfies $\mathrm{Out}(T) = \langle N, N, N\rangle$, and by Theorem 23 every colored tensor $Z \in \mathcal{I}(T)$ has $\chi(Z) = 1$ (since it is a product of three colored tensors with chromatic number 1).

By Theorem 18, $\chi(\langle N, N, N\rangle) \le N^{1+o(1)}$, and finally using Theorem 21,

$$\chi(T) \le \chi(\mathrm{Out}(T)) \max_{T' \in \mathcal{I}(T)} \chi(T') \le N^{1+o(1)}.$$

The communication complexity bound now follows from the characterization in Theorem 9. □

3.6 Zeroing out subrank of the matrix multiplication tensor

We will use the connection between number on the forehead communication complexity, and structural properties of matrix multiplication tensor to find a non-trivial *upper bound* for the zero-out subrank of matrix multiplication tensor — i.e., an upper bound on the size of the largest diagonal subtensor to which matrix multiplication tensor can be zeroed-out.

Authors of the work [LPS19] observed a connection between the NOF communication complexity and the Ruzsa–Szemerédi problem. In particular, they proved a super-constant lower bound for NOF communication complexity of *any* permutation problem, as a corollary of the Ruzsa–Szemerédi theorem (which in turns is proved via the triangle-removal lemma).

Here, we observe that with our connection between NOF communication and the matrix-multiplication tensor, we can use their proof to show a somewhat stronger statement — the upper bound on the zero-out subrank of matrix multiplication.

Theorem 25. *The zero-out subrank of matrix multiplication satisfies*

$$Q_{zo}(\langle N, N, N\rangle)) \leq N^2/2^{c\log^* N}.$$

Note that this theorem (together with Theorem 27) implies in fact the $\Omega(\log^* N)$ lower bound for all NOF permutation problems present in [LPS19], and the proof is almost identical.

Before we continue, we shall state a convenient well-known equivalent formulation of the Ruzsa–Szemerédi theorem. The sub-quadratic upper bound of the quantity below has been famously proven by Ruzsa and Szemerédi in 1978 [RS78], using the notorious triangle removel lemma (in fact they considered a slightly different formulation, the $(6, 3)$-problem, but it can be easily seen to be equivalent to the following statement [CEMS91]). Substituting a quantitative version of the triangle removal lemma by Fox [Fox11] in their proof leads to the following quantitative bound.

Theorem 26 ([RS78, CEMS91, Fox11])**.** *For any graph $G(V, E)$, let $\mathcal{C}$ be a collection of all triangles in the graph G. If all those triangles are edge-disjoint, then $|\mathcal{C}| \leq N^2/2^{c\log^* N}$ where $N = |V|$.*

With this theorem in hand, it is easy to show a sub-quadratic upper bound on the largest independent set in the matrix multiplication tensor. In fact the quantitative question about the density of the largest independent set in the matrix multiplication tensor is equivalent to the Ruzsa–Szemerédi problem.

Proof of Theorem 25. Consider an independent set $\tilde{S} \subset \langle N, N, N\rangle$ — we can treat it as a subset $S \subset [N] \times [N] \times [N]$. Let us now consider the following tri-partite graph on $[N] \times [3]$: for each $(a, b, c) \in S$ we put a triangle on vertices $(a, 0), (b, 1)$ and $(c, 2)$. This construction yields a graph G on $3N$ vertices, together with a collection $\mathcal{C}$ of its $|S|$ triangles. To appeal to the Theorem 26, we need to show that triangles in the collection $\mathcal{C}$ are edge-disjoint, and that those are all the triangles in a graph G.

First of all, those are edge disjoint — if we had a pair of distinct triangles in $\mathcal{C}$ sharing an edge, it would correspond to two elements $(a, b, c_1) \in S$ and

$(a, b, c_2) \in S$ — a contradiction with an assumption that S is an independent set in matrix multiplication tensor.

Similarly, since the graph G is tri-partite, the only possible triangle are of form $(a, 0), (b, 1), (c, 2)$ for some a, b, c. We wish to show that if S was an independent set, this is possible only for $(a', b', c') \in S$.

Indeed, if the edge $(a, 0), (b, 1)$ was included in the graph, there must have been a tuple $(a, b, c') \in S$, and analogously $(a, b', c) \in S, (a', b, c) \in S$ for some a', b', c'. Since S was an independent set, this readily implies that $a' = a, b' = b, c' = c$ and indeed $(a, b, c) \in S$. □

3.6.1 Lower bound for matrix multiplication subrank from NOF protocols

In the study of the matrix multiplication tensor, lower bounds on its zero-out subrank are much more consequential than upper bounds. Strassen proved that $\alpha(\langle N, N, N\rangle) \geq N^{2-o(1)}$, which is almost maximal (since the tensor $\langle N, N, N\rangle$ has size $N^2 \times N^2 \times N^2$, the largest diagonal tensor one could potentially hope to find there is of size N^2). He used this together with the Laser method to design a fast matrix multiplication algorithm in [Str86], and all subsequent record-holding matrix multiplication algorithms have also used this. At the same time, recent barrier results [AW21, Alm21, CVZ21] used the lower bounds on the subrank of matrix multiplication to prove barrier against certain approaches for proving that the matrix multiplication constant $\omega = 2$. (Roughly speaking, using our notation here, they show that since $\alpha(\langle N, N, N\rangle)$ is so large, one would need to use an intermediate tensor T with $\alpha(T)$ also large in order to design a sufficiently fast algorithm.)

One of the crucial ingredients in the proof using the Laser method that $\alpha(\langle N, N, N\rangle) \geq N^{2-o(1)}$ is leveraging the existence of a subset $S \subset [N]$ which does not have three-term arithmetic progressions, and has relatively high density. Using Behrend's [Beh46] construction of such a set with density $2^{-\mathcal{O}(\sqrt{\log N})}$, it follows that $\alpha(\langle N, N, N\rangle) \geq N^2/2^{\mathcal{O}(\sqrt{\log N})}$.

Using the same construction of Behrend's set, it was proved in [CFL83] that the NOF problem $\textsc{Eval}(\mathbb{Z}_N)$ has a deterministic protocol with communication complexity $\mathcal{O}(\sqrt{\log N})$ — significantly improving upon the naive $\mathcal{O}(\log N)$. We observe that their construction of an efficient NOF protocol not only uses the same technical ingredients as the construction of large independent set in the matrix multiplication tensor, but in fact it is much more intimately related — known results in communication complexity together with our connection directly imply the Strassen result on the subrank of matrix multiplication tensor.

Proposition 27. *If there is any permutation problem $I \subset [N] \times [N] \times [N]$ with NOF communication complexity $\mathcal{CC}(\tilde{I}, P_{\text{NOF}}) \leq k$, then $\alpha(\langle N, N, N\rangle) \geq N^2 2^{-k}$.*

Proof. By Theorem 9, if the promise problem $(\tilde{I}, P_{\text{NOF}})$ has NOF communication complexity k, then $\chi(I, P_{\text{NOF}}) \leq 2^k$, and in particular taking the largest color we get $\alpha(I, P_{\text{NOF}}) \geq n^2 2^{-k}$. Since the promise tensor P_{NOF} for the number on the forehead communication problems is exactly the matrix multiplication tensor $\langle n, n, n\rangle$, we can disregard which terms are green, and deduce $\alpha(\langle n, n, n\rangle) \geq \alpha(I, P_{\text{NOF}}) \geq n^2 2^{-k}$. □

3.7 Other promise problems

As we understand now, the slice rank-based approach for proving lower bounds for asymptotic communication complexity of NOF permutation problems failed, because the technique depends only on the promise tensor, and is insensitive to the specific problem — and the promise tensor corresponding to the number-on-the-forehead problems is a matrix multiplication tensor, with maximal subrank.

With this in mind, it is worth looking at promise problems with other promise-tensors, hopefully ones for which we know the asymptotic subrank is not maximal — this statement alone shall imply that *every* permutation problem with such a promise should have positive asymptotic communication complexity.

Fact 28. *If the promise tensor P over $\Sigma \times \Sigma \times \Sigma$ has non-maximal asymptotic subrank*

$$\underline{\alpha}(P) := \limsup \alpha(P^{\otimes N})^{1/N} < |\Sigma|,$$

then for every permutation problem $I \subset P$ we have

$$\mathcal{CC}((I,P)^{\otimes N}) \geq \Omega(N).$$

The proof of this fact is straightforward with what we have discussed so far. We later provide a complete proof of a strictly stronger theorem (Theorem 33), so we leave a direct proof of this special case to the reader.

The solution of the cap-set problem [CLP17, EG16, Tao16] and its subsequent generalizations [BCC$^+$17], can be interpreted as saying that for any abelian group G, the structure tensor

$$T_G := \sum_{g,h \in G} x_g y_h z_{-g-h},$$

has $\underline{\alpha}(T_G) < |G|$. In particular, all permutation problems with a promise tensor being a structure tensor of such a group, have positive asymptotic communication complexity. For example, the following is true

Corollary 29. *Consider the promise problem $EQ_{\mathbb{F}_3^N}$, where parties receive inputs $x, y, z \in \mathbb{F}_3^N$ with a promise that $x + y + z = 0$, and they wish to decide whether $x = y$. This problem requires $\Omega(N)$ communication.*

Without the preceding discussion it would not be immediately clear whether we should expect this statement to be true — the relevant part of the problem is between Alice and Bob, and they just wish to decide the equality predicate for their inputs — a task which, with ordinary deterministic two-party protocols, requires $\Omega(N)$ communication. But should we expect Charlie, who knows $x + y$ to be of much help in solving the task?

On the other hand, one might think that the statement above — if true — should be relatively straightforward to prove directly, with a proof similar to the communication lower bound for deterministic two-party equality problem. We observe that in fact this statement is easily *equivalent* to the cap-set theorem, hence a direct proof would be an interesting new proof of what used to be a cap-set *conjecture* for almost four decades.

Fact 30. *For any abelian group G, if there is a $S \subset G$ of size $|G|^{1-\delta}$ without three-terms arthmetic progressions, then $\mathcal{CC}(EQ_G) \leq \delta \log |G| + \mathcal{O}(\log \log |G|)$*

Proof. The corresponding communication tensor of a problem EQ_G is (I, T_G) where T_G is the structure tensor of the group G, and $I = \{(x, x, -2x) : x \in G\} \subset T_G$.

We will first show that $\alpha(I, T_G) \geq |S|$. We wish to show that if S does not have three terms arithmetic progressions, then $S' := \{(x, x, -2x) : x \in S\} \subset I$ is an independent set. Indeed, for the sake of contradiction, let us assume that $(x, y, -x - y)$ for some $x \neq y$ has $x \in \pi_1(S'), y \in \pi_2(S'), -x - y \in \pi_3(S')$. This means $x, y \in S$, and $-x - y = -2z$ for some $z \in S$ — i.e. x, z, y is a three terms arithmetic progression.

To lift a large independent set to a small coloring, we will refer to Theorem 17 — it is enough to show that there is an automorphism of (I, T_G) that maps arbitrary $(x, x, -2x)$ to $(y, y, -2y)$. This is given by a triple of permutations $\sigma_1 = \sigma_2 : w \mapsto w - x + y$, and $\sigma_3 : w \mapsto w + 2y - 2x$. Finally, by Theorem 9 a coloring of the tensor (I, T_G) into $|G|^\delta \log |G|$ colors yields a protocol with complexity $\delta \log |G| + \log \log |G|$. □

3.8 Lower Bounds from the Asymptotic Spectrum of Tensors

The results in this section concern promise problems which are not necessarily permutation problems. We will first need a simple combinatorial description of the communication complexity of such a problem.

For any transcript of a protocol π, the set of inputs $S_\pi \subset \Sigma \times \Sigma \times \Sigma$ which result in such a transcript is a combinatorial cube $A_\pi \times B_\pi \times C_\pi$ for some $A_\pi, B_\pi, C_\pi \subset \Sigma$. If the protocol is correct, in every such cube either all terms from P are contained in I, or they all are not in I. As such, a correct protocol of complexity k induces a partition of the tensor P into 2^k monochromatic combinatorial cubes (where a cube is monochromatic in the preceding sense).

In particular, for any promise problems (I, P) all terms from I can be covered by $2^{\mathcal{CC}(I,P)}$ combinatorial cubes, each of which does not contain any term in $P - I$.

Theorem 31. *Let $r : \mathbb{F}^{\Sigma \times \Sigma \times \Sigma} \to \mathbb{R}_{\geq 0}$ be any function which is:*

- sub-additive*, meaning, if $A, B \in \mathbb{F}^{\Sigma \times \Sigma \times \Sigma}$ then $r(A + B) \leq r(A) + r(B)$, and*
- monotone under zeroing outs*, meaning, if $A, B \in \mathbb{F}^{\Sigma \times \Sigma \times \Sigma}$ and B is a zeroing out of A, then $r(B) \leq r(A)$.*

Suppose P is any promise tensor over $\Sigma \times \Sigma \times \Sigma$, and $I \subset P$ is any problem (not necessarily a permutation problem). Then,

$$\mathcal{CC}(I, P) \geq \log_2 \left(\frac{r(I)}{r(P)} \right).$$

Proof. Let $k = 2^{\mathcal{CC}(I,P)}$. By the preceding discussion, there are k tensors $T_1, \ldots, T_k \in \mathbb{F}^{\Sigma \times \Sigma \times \Sigma}$ which are each zeroing outs of P, such that $T_1 + \cdots + T_k = I$. We thus have

$$r(I) \leq \sum_{i=1}^{k} r(T_i) \leq k \cdot r(P),$$

as desired. □

Although it is relatively simple, Theorem 31 is quite powerful since there are many well-studied examples of functions r in algebraic complexity and tensor combinatorics which are sub-additive and monotone under zeroing outs. Tensor rank (R) is the most prominent example, but there are also other well-studied rank variants such as 'slice rank' [Tao16] and 'geometric rank' [KMZ20]. (Note that subrank (Q) is super-additive, not sub-additive.)

Number In Hand For one example, consider the all-1s tensor $P_{NIH} \in \mathbb{F}^{\Sigma\times\Sigma\times\Sigma}$, which is the promise tensor for the Number In Hand model. This very simple tensor has rank $R(P_{NIH}) = 1$. Thus, by Theorem 31, any tensor I with $R(I) \geq |\Sigma|^{\Omega(1)}$ has essentially maximal Number In Hand communication complexity $\Omega(\log|\Sigma|)$. Note that all tensors $I \in \mathbb{F}^{\Sigma\times\Sigma\times\Sigma}$ have $R(I) \geq |\Sigma|$ except for tensors which are somehow "degenerate"; for instance, all 'concise' tensors have this property [CGL$^+$21].

This instantiation of Theorem 31, where the promise is the all-ones tensor and r is the rank of the problem tensor I, is akin to the standard lower bound in two-party deterministic communication complexity by the log rank of the communication matrix. The celebrated log-rank conjecture stipulates that in the standard setting this statement can be weakly inverted, and in fact any two-party communication problem (with trivial promise) has a protocol with complexity $(\log \operatorname{rank}(I))^c$ for some constant c.

Number On Forehead For another example, consider the tensor $P_{NOF} \in \mathbb{F}^{\Sigma^2\times\Sigma^2\times\Sigma^2}$, which is the promise tensor for the Number On Forehead model. As we've seen, $P_{NOF} = \langle|\Sigma|, |\Sigma|, |\Sigma|\rangle$, and so we know that $R(P_{NOF}) = |\Sigma|^{\omega+o(1)}$. It follows that if $I \subset P_{NOF}$ is a problem with $R(I) \geq |\Sigma|^c$ for some constant $c > \omega$, then I has Number On Forehead communication complexity $\geq (c - \omega + o(1))\log|\Sigma|$.

This relates the classic open problem in algebraic complexity of finding high-rank tensors to the task of proving deterministic NOF lower bounds. Furthermore, the fact that ω, the exponent of matrix multiplication, is *subtracted* in the communication lower bound is intriguing. This means that designing faster matrix multiplication algorithms, and hence decreasing our best-known upper bound on ω, actually makes it *easier* to prove NOF *lower bounds* via this approach.

3.8.1 Generalization to the Asymptotic Spectrum

Finally, in this section, we note that while subrank is not sub-additive, the zeroing out variant (Q_{ZO}) we have been studying is additive (and hence sub-additive) for direct sums, and so a similar version of Theorem 31 still holds for it. Moreover, this holds for any r in the 'asymptotic spectrum of tensors' [Str86, WZ22], a well-studied class which roughly captures the different ways to generalize the notion of the 'rank' of a matrix to tensors.

Definition 32. *Tensor B is an* identification *of tensor A if you can get to B from A by first doing a zeroing out then setting some variables equal to each other. For example, $B = x_0y_0z_0 + x_0y_1z_1$ is an identification of $A = x_0y_0z_0 + x_1y_1z_1 + x_0y_1z_2$ by zeroing out z_2 and setting $x_1 \leftarrow x_0$. Identifications generalize zeroing outs, but are a special case of tensor restrictions.*

Theorem 33. *Let $r : \mathbb{F}^{\Sigma\times\Sigma\times\Sigma} \to \mathbb{R}_{\geq 0}$ be any function which is:*

- sub-additive for direct sums*, meaning, if $A \in \mathbb{F}^{\Sigma\times\Sigma\times\Sigma}$ and $B \in \mathbb{F}^{\Sigma'\times\Sigma'\times\Sigma'}$ are tensors for disjoint sets Σ, Σ', then $r(A+B) \geq r(A) + r(B)$, where $A+B$ is a tensor over $\mathbb{F}^{\Sigma\cup\Sigma'\times\Sigma\cup\Sigma'\times\Sigma\cup\Sigma'}$ and*
- monotone under identifications*, meaning, if $A, B \in \mathbb{F}^{\Sigma\times\Sigma\times\Sigma}$ and B is an identification of A, then $r(B) \leq r(A)$.*

Suppose P is any promise tensor over $\Sigma\times\Sigma\times\Sigma$, and $I \subset P$ is any problem (not necessarily a permutation problem). Then,

$$\mathcal{CC}(I, P) \geq \log_2\left(\frac{r(I)}{r(P)}\right).$$

Proof. Let $k = 2^{\mathcal{CC}(I,P)}$. By definition, there are k tensors $T_1, \ldots, T_k \in \mathbb{F}^{\Sigma\times\Sigma\times\Sigma}$ which are each zeroing outs of P, such that $T_1 + \cdots + T_k = I$. In particular, the direct sum of k copies of P has a zeroing out to the direct sum of all the T_i tensors, which in turn has an identification to I. We thus have

$$r(I) \leq \sum_{i=1}^{k} r(T_i) \leq k \cdot r(P),$$

as desired. □

Fact 34. *The zeroing out subrank $r = Q_{ZO}$ satisfies the premises of Theorem 33.*

Proof. Q_{ZO} is sub-additive for direct sums since any zeroing out of a direct sum to a diagonal tensor must separately zero out each part into a diagonal tensor. It's monotone under identifications since, when transforming a tensor A to a diagonal tensor via identification, we may assume that we first perform a zeroing out and then set variables equal to each other, but setting variables equal to each other cannot increase the size of a diagonal tensor. □

3.9 Intermediate Group Promise Problems

Suppose $T_1 \subset T_2 \subset T_3$ are $\{0,1\}$-valued tensors. These give rise to three promise problems, depending on which one picks as the promise and which one picks as the problem: (T_1, T_2), (T_1, T_3), and (T_2, T_3). We observe a simple triangle inequality-type relationship between their communication complexities:

Fact 35. $\mathcal{CC}(T_1, T_3) \leq \mathcal{CC}(T_1, T_2) + \mathcal{CC}(T_2, T_3)$

Proof. In order to solve (T_1, T_3), first use a protocol for (T_2, T_3) to determine whether the input is in T_2, then use a protocol for (T_1, T_2) to determine whether the input is in T_1. □

We will now consider a particular instantiation of Theorem 35 which yields a potential avenue toward proving deterministic NOF lower bounds. T_1, T_2, T_3 will be tensors from $\mathbb{F}^{\Sigma^2\times\Sigma^2\times\Sigma^2}$ for some finite set Σ. Let $n := |\Sigma|$.

We will pick T_3 to be the structure tensor of the group $(\mathbb{Z}/n\mathbb{Z})^2$. This tensor is typically written

$$T_3 = \sum_{a,b,c,d\in[n]} x_{(a,b)} y_{(c,d)} z_{(a+c,b+d)},$$

where additions in the subscripts (here and in the remainder of this subsection) are done mod n. However, to simplify the next step, we can permute the variable names (replacing $x_{(a,b)}$ with $x_{(a+b,b)}$ for all $a, b \in [n]$, replacing $y_{(c,d)}$ with $y_{(c,c+d)}$ for all $c, d \in [n]$, and replacing $z_{(a+c,b+d)}$ with $z_{(b+d,a+c)}$ for all $a, b, c, d \in [n]$) to equivalently write it as

$$T_3 = \sum_{a,b,c,d \in [n]} x_{(a+b,b)} y_{(c,d+c)} z_{(b+d,a+c)}.$$

Next, we will pick $T_2 = P_{\text{NOF}} \in \mathbb{F}^{n^2 \times n^2 \times n^2}$ to be the Number On Forehead promise tensor, i.e., the $\langle n, n, n \rangle$ matrix multiplication tensor,

$$T_2 = \sum_{i,j,k \in [n]} x_{(i,j)} y_{(j,k)} z_{(k,i)}.$$

Fact 36. $T_2 \subset T_3$.

Proof. For each $i, j, k \in [n]$, we need to prove that there is a choice of $a, b, c, d \in [n]$ such that $x_{(i,j)} y_{(j,k)} z_{(k,i)} = x_{(a+b,b)} y_{(c,d+c)} z_{(b+d,a+c)}$. The choice which achieves this is $b = c = j$, $a = i - j \pmod{n}$, and $d = k - j \pmod{n}$. □

Finally, we may pick T_1 to be any Number On Forehead problem $T_1 \subset T_2$. Theorem 35 now relates the Number On Forehead complexity of the problem T_1 to two different promise problems with the promise T_3:

$$\mathcal{CC}(T_1, P_{NOF}) \geq \mathcal{CC}(T_1, T_3) - \mathcal{CC}(P_{NOF}, T_3).$$

Using the tools we've developed thusfar, we can bound the communication complexity $\mathcal{CC}(P_{NOF}, T_3)$. Note that these are tensors over $\mathbb{F}^{n^2 \times n^2 \times n^2}$, so the trivial communication upper bound would be $2 \log n$.

Lemma 37.

$$(\omega - 2) \log n \leq \mathcal{CC}(P_{NOF}, T_3) \leq \log(n).$$

Proof. Since P_{NOF} is the $\langle n, n, n \rangle$ matrix multiplication tensor, its rank is $R(P_{NOF}) = n^{\omega + o(1)}$. Meanwhile, since T_3 is the structure tensor of an abelian group, its rank is $R(T_3) = n^2$ (see, e.g., [CU03, Theorem 2.3 and Theorem 4.1]). Thus, Theorem 31 yields the lower bound $(\omega - 2) \log n \leq \mathcal{CC}(P_{NOF}, T_3)$.

For the upper bound, we directly give a communication protocol. In this problem, player A is given $(a + b, b)$, player B is given $(c, d + c)$, and player C is given $(b + d, a + c)$, for some values $a, b, c, d \in \mathbb{Z}_n$, and their goal is to determine whether or not the following three equalities hold: $b = c$, $d + c = b + d$, and $a + c = a + b$. Note that this is equivalent to testing whether $b = c$, since the other two equalities follow from this. They can do this by having player A send b, which uses $\log n$ bits, and then having player B confirm that it is equal to c. □

Interestingly, in light of the lower bound in Theorem 37, we know that improving the upper bound $\mathcal{CC}(P_{NOF}, T_3) \leq \log(n)$ *requires* using fast matrix multiplication. Indeed, any bound $\mathcal{CC}(P_{NOF}, T_3) \leq (1 - \delta) \log(n)$ would imply that $\omega < 3 - \delta$. More specifically, one can observe that an upper bound on ω obtained in this way would fall under the 'group-theoretic approach' to matrix

multiplication [CU03], using the group $(\mathbb{Z}/n\mathbb{Z})^2$. There are known barriers to this approach for any fixed n [BCC+17], but it's consistent with the best-known barriers that this approach could prove $\omega = 2$ by using increasingly larger n. (On the other hand, most recent work on the group-theoretic approach has focused instead on non-abelian groups [BCG+22].)

Combining Theorem 37 together with Theorem 35 shows that

Fact 38.

$$\mathcal{CC}(T_1, P_{NOF}) \geq \mathcal{CC}(T_1, T_3) - \log(n).$$

For any T_1, the straightforward algorithm for the problem (T_1, T_3) has communication complexity $2\log(n)$. Theorem 38 shows that any T_1 with $\mathcal{CC}(T_1, T_3) \geq (1+\delta)\log(n)$ for any fixed $\delta > 0$ would give a linear Number On Forehead lower bound $\mathcal{CC}(T_1, P_{NOF}) \geq \Omega(\log n)$.

In light of this, improving the upper bound of Theorem 37 could be quite powerful. In particular, if $\omega = 2$, and if this can be used to give an algorithm achieving $\mathcal{CC}(P_{NOF}, T_3) \leq o(\log(n))$, then any T_1 with $\mathcal{CC}(T_1, T_3) \geq \Omega(\log(n))$ (without any restriction on the leading constant) would give a linear Number On Forehead lower bound $\mathcal{CC}(T_1, P_{NOF}) \geq \Omega(\log n)$ as well.

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